

Day 4 – Attempted Code Explanation

Line	Explanation
<code>import cv2</code>	Access camera and draw landmarks using OpenCV.
<code>from mediapipe.tasks.python.vision import HandLandmarker, HandLandmarkerOptions</code>	Imports classes for hand detection.
<code>from mediapipe.tasks.python import BaseOptions</code>	Base options for model path.
<code>import time</code>	For timestamp calculation per frame.
<code>options = HandLandmarkerOptions(...)</code>	Initialize hand landmarker with model and hand count.
<code>landmarker = HandLandmarker.create_from_options(options)</code>	Creates the landmarker object.
<code>cap = cv2.VideoCapture(0)</code>	Opens default camera.
<code>start_time = int(time.time() * 1000)</code>	Records initial timestamp.
<code>frame = cv2.flip(frame, 1)</code>	Flip frame for mirror view.
<code>frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)</code>	Converts frame to RGB.
<code>timestamp_ms = int(time.time() * 1000) - start_time</code>	Frame timestamp for <code>detect_for_video</code> .
<code>results = landmarker.detect_for_video(frame_rgb, timestamp_ms)</code>	Attempts hand landmark detection.
<code>cv2.circle(frame, (x, y), 5, (0, 255, 0), -1)</code>	Draw green dots for each landmark.
<code>cv2.imshow(...)</code>	Displays camera feed.
<code>cv2.waitKey(1) & 0xFF == 27</code>	Stops loop if ESC pressed.

Why This Code Did Not Work

- Tasks API requires MediaPipe Image objects, which are missing in Python 3.12 version.
- Feeding a NumPy array triggers `_image_ptr` error.
- Despite correct model file and code structure, no landmarks could be collected.
- Chose to continue with Python 3.12 as a temporary workaround, but this prevents full functionality.

So here in day four the same as day three there is code failure but we learned a lot today.